Differential Privacy in Data Analytics

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1. Abstract

Differential privacy (DP) is a formal framework for limiting the amount of information that can be inferred about any single individual from released statistics. In this project, we compare two widely used DP mechanisms, the Laplace mechanism and the Gaussian mechanism: on common data analytics tasks using the Adult Income dataset. We evaluate three aggregate queries: a count query, a mean query, and a histogram query. Utility is measured using mean absolute error and mean squared error across multiple privacy budgets. The results show the expected privacy-utility trade-off: stronger privacy settings introduce more distortion, while larger privacy budgets improve accuracy. Across all tested queries and privacy budgets, the Laplace mechanism produced lower error than the Gaussian mechanism, providing a better privacy-utility balance for the low-sensitivity aggregate statistics considered in this project.

2. Introduction

Frequently, organizations wish to output useful statistical information while preserving the privacy of individuals whose information is included in the data set. Some examples of such scenarios include universities releasing information about their students, hospitals releasing demographic information, and corporations releasing information about their workforce. However, it is often the case that traditional anonymization is insufficient, since aggregate information released can be used to infer sensitive information when external information is available **[1], [8]**. Differential privacy solves this problem since it guarantees that the output of a computation will not significantly change when a single individual is added or removed from the data set **[1]**.

This report is a project in the realm of basic data analytics and differential privacy. The project aims to experimentally compare the Laplace mechanism for providing pure ε-differential privacy and the Gaussian mechanism for providing approximate (ε, δ)-differential privacy **[1], [3]**. These mechanisms will be compared for three basic forms of aggregate queries: counting, mean, and histogram.

The main objectives of the project are:

- To implement Laplace and Gaussian noise addition for aggregate queries.
- To evaluate how accuracy changes as the privacy budget changes.

- To compare the behavior of the two mechanisms across different query types.
- To discuss the practical considerations involved in selecting a DP mechanism for real-world analytics.

In our use case, we have a university/institution that wants to release anonymized statistics but does not want to disclose any information on any particular individual. Differential privacy provides a framework for such a case.

3. Detailed Description of the Privacy-Enhancing Techniques, Problem, and Development

3.1 Problem Setting

The study discussed here deals with the privacy-preserving dissemination of statistical summaries of data. The Adult Income data set from the UCI Machine Learning Repository is used **[9]**. After removing the rows with missing values, the data set is clean and ready for use. It has 30,162 instances and 14 attributes. The research focuses only on the attributes of the data set that are relevant to straightforward analytics. These are age, hours per week, income category, and education level.

The queries are of the following types:

- Count query: the count of individuals whose income is above $50K.
- Mean query: the mean age of the individuals.
- Histogram query: the frequency of education levels.

3.2 Differential Privacy Background

A randomized algorithm M satisfies ε-differential privacy if for any two neighboring datasets D and D` that differ in one individual, and for any set of possible outputs S,

$$\Pr[M(D) \in S] \leq e^{\varepsilon} \Pr[M(D`) \in S$$

This definition guarantees that the presence or absence of one individual has limited influence on the released result. Smaller values of ε provide stronger privacy but usually reduce utility because more noise must be added **[1]**.

The Gaussian mechanism instead provides (ε, δ)-differential privacy, which relaxes the guarantee slightly by allowing a small failure probability δ **[3], [4]**. In this project, δ is fixed at 10^{-5}.

3.3 Sensitivity Analysis

A core requirement of differential privacy is sensitivity calibration [1]. The amount of noise added depends on how much a query can change when one record changes.

- **Count query sensitivity:** $\Delta f = 1$, because adding or removing one record changes the count by at most one.
- **Mean age sensitivity:** for a bounded attribute, the sensitivity is

$$\Delta f = \frac{max(age) - min(age)}{n}$$

  This bounds the maximum influence of one individual on the mean.

- **Histogram sensitivity:** under the add/remove model, one individual contributes to exactly one bin, so the histogram sensitivity is 1 in L_1-norm.

Correct sensitivity selection is critical. If sensitivity is underestimated, the privacy guarantee becomes invalid. If it is overestimated, too much noise may be added, harming utility **[1], [5]**.

3.4 Laplace Mechanism

The Laplace mechanism achieves pure ε-differential privacy by adding noise drawn from a Laplace distribution **[1]**:

$$M(D) = f(D) + Lap\left(\frac{\Delta f}{\varepsilon}\right)$$

This mechanism is natural for numeric queries with known L_1-sensitivity **[1]**. In our project, it is applied to the count query, mean query, and histogram bins. The implementation uses Python with NumPy to generate Laplace noise and pandas to load and preprocess the dataset.

A major advantage of the Laplace mechanism is its conceptual simplicity. It is easy to calibrate for low-dimensional statistics and directly satisfies pure ε-DP. However, as ε becomes small, the added noise can become large enough to significantly distort the result.

3.5 Gaussian Mechanism

The Gaussian mechanism adds Gaussian noise to the true answer of the query and achieves (ε, δ) -differential privacy as follows **[3]**:

$$M(D) = f(D) + \mathcal{N}(0, \sigma^2).$$

The choice of the noise scale is dependent on the query sensitivity, the privacy budget, and δ. The mechanism is particularly beneficial when the processes are repeated because the privacy loss of the mechanism is easier to handle naturally under advanced composition and Rényi differential privacy **[4]**.

In the project, the Gaussian mechanism is used for the same queries as the Laplace noise mechanism. Therefore, the results of the pure differential privacy approach are compared to the results of the approximate differential privacy approach under the same conditions.

3.6 Experimental Setup

All experiments are implemented in Python 3. The dataset is preprocessed using pandas, and random noise is generated with NumPy. We evaluate both mechanisms using privacy budgets:

$$\varepsilon \in \{0.1,\ 0.5,\ 1.0,\ 2.0\}.$$

For the Gaussian mechanism, we fix:

$$\delta = 10^{-5}.$$

For each query and each privacy budget, the privatized answer is compared against the true answer. The main utility metrics are:

- **Absolute Error:** $|\hat{f}(D) - f(D)|$
- **Mean Squared Error (MSE):** average squared deviation over repeated trials

To make the comparison reliable, each mechanism should be run multiple times for every ε value and query type, and the final report should summarize either the average error or the distribution of errors **[5]**.

The implementation, result files, and supporting materials are available in our project repository

3.7 Results and Discussion

After preprocessing the Adult Income data set by removing the rows with missing values, the data set contained 30,162 data points. The non-private baseline statistics showed the number of individuals with income levels above $50K as 7,508, the number of individuals with income levels at or below $50K as 22,654, the mean age of the individuals as 38.4379 years, and the average hours of work per week as 40.9312. These values are the baseline values to compare the privatized values against.

The experiments are designed based on the methodology provided in the framework of the project. The experiments are designed for both Laplace and Gaussian mechanisms for the values of ε = 0.1, 0.5, 1.0, 2.0. In the case of the Gaussian mechanism, δ is set to 10^-5.

3.7.1 Count Query Results

Analysis of the count query reveals that the Laplace mechanism consistently outperforms the Gaussian mechanism across the entire spectrum of privacy parameters. Under a restrictive privacy budget of ε =

0.1, the Laplace mechanism maintains a mean absolute deviation slightly above 9.1 and a mean squared error of less than 151. In comparison, the Gaussian mechanism under the same constraints produces an average error exceeding 37 and a mean squared error surpassing 2418. The observed increase in accuracy as the privacy budget expands underscores the fundamental trade-off between individual privacy and statistical utility. Specifically, at ε = 2.0, the Laplace mechanism achieves a mean absolute error of approximately 0.56, whereas the Gaussian mechanism yields a higher error rate of approximately 2.05.

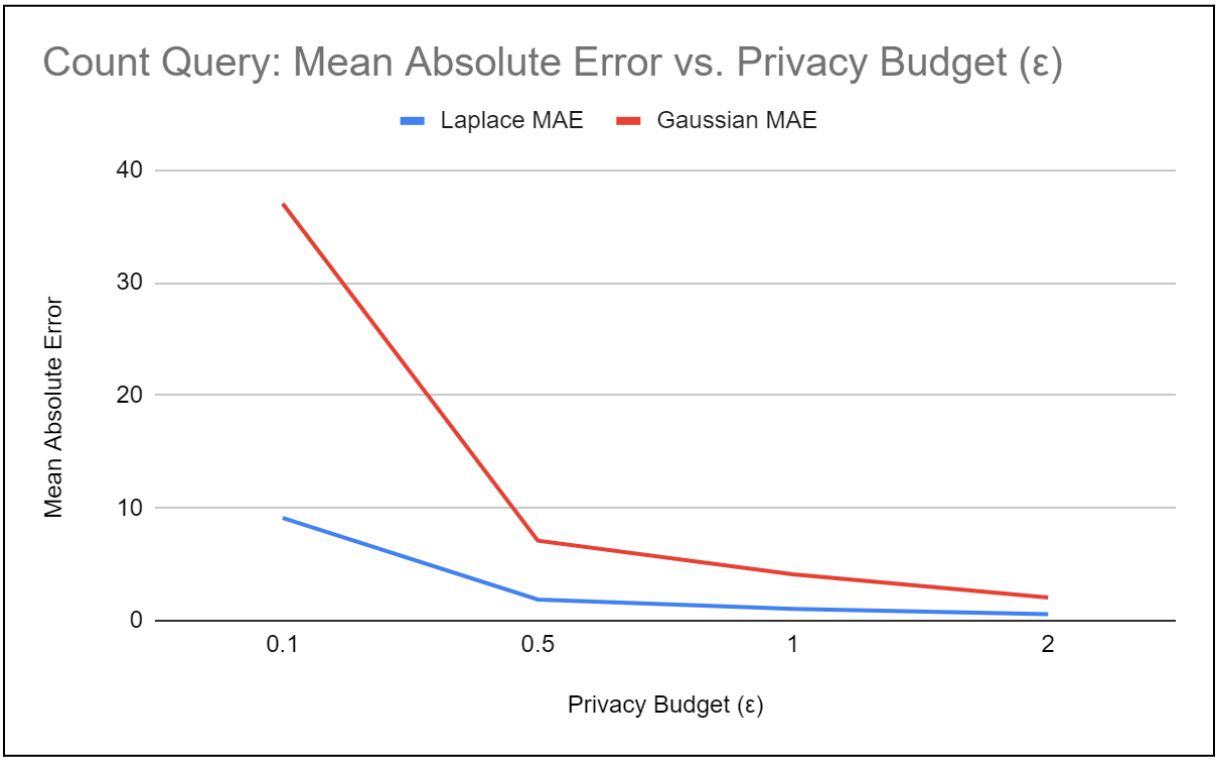

Figure 1. Mean absolute error for the count query under the Laplace and Gaussian mechanisms across different privacy budgets. As ε increases, the error decreases for both mechanisms, and the Laplace mechanism consistently produces lower error than the Gaussian mechanism.

3.7.2 Mean Query Results

The evaluation of age-related mean queries demonstrated consistent performance from both the Laplace and Gaussian mechanisms, primarily due to the low sensitivity of the bounded mean relative to the dataset size. However, the Laplace mechanism consistently exhibited lower error rates across all tested ε values. At ε = 0.1, the Laplace mechanism recorded an error of 0.0235, compared to 0.0981 for the Gaussian mechanism. This performance gap persisted at ε = 2.0, where the errors were 0.0013 and 0.0045, respectively. These results suggest that mean queries are particularly resilient to noise injection, as the sensitivity is mitigated by the total sample volume.

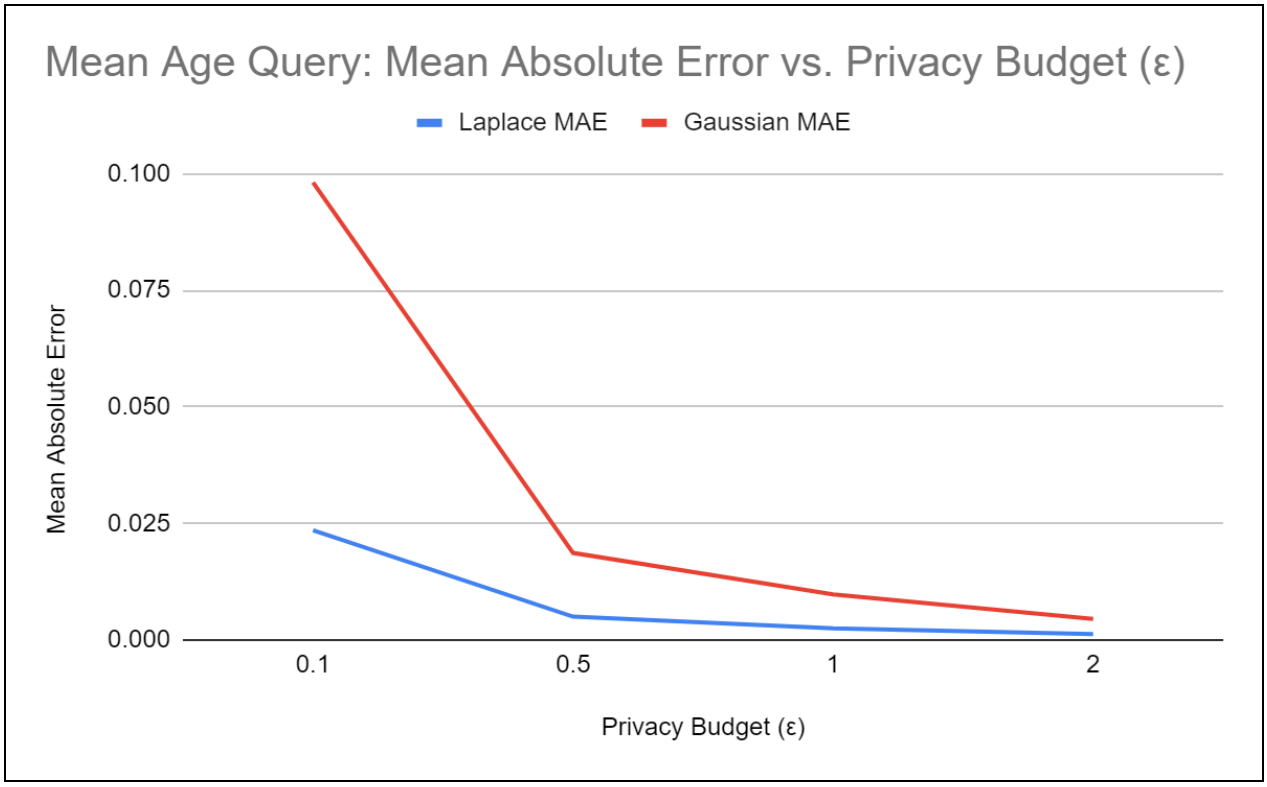

Figure 2. Mean absolute error for the mean age query under the Laplace and Gaussian mechanisms across different privacy budgets. As ε increases, the error decreases for both mechanisms. The Laplace mechanism again produces lower error than the Gaussian mechanism, although the difference is smaller because the bounded mean has very low sensitivity.

3.7.3 Histogram Query Results

Unexpectedly large gaps appeared in the histogram when comparing education level outputs across methods. Because multiple bins received added noise, the histogram query necessitated a dilution of the allocated privacy budget. At ε = 0.1, error measures climbed sharply under Gaussian noise - mean absolute deviation reached 38.6114, mean squared error hit 2330.1312. In contrast, Laplace produced smaller mistakes: absolute deviations averaged 9.8272, squared errors came in at 195.5275. When privacy relaxed to ε = 2.0, Laplace still outperformed clearly, posting a modest 0.5037 MAE. Meanwhile, its counterpart lagged behind, recording nearly four times that value at 1.9694.

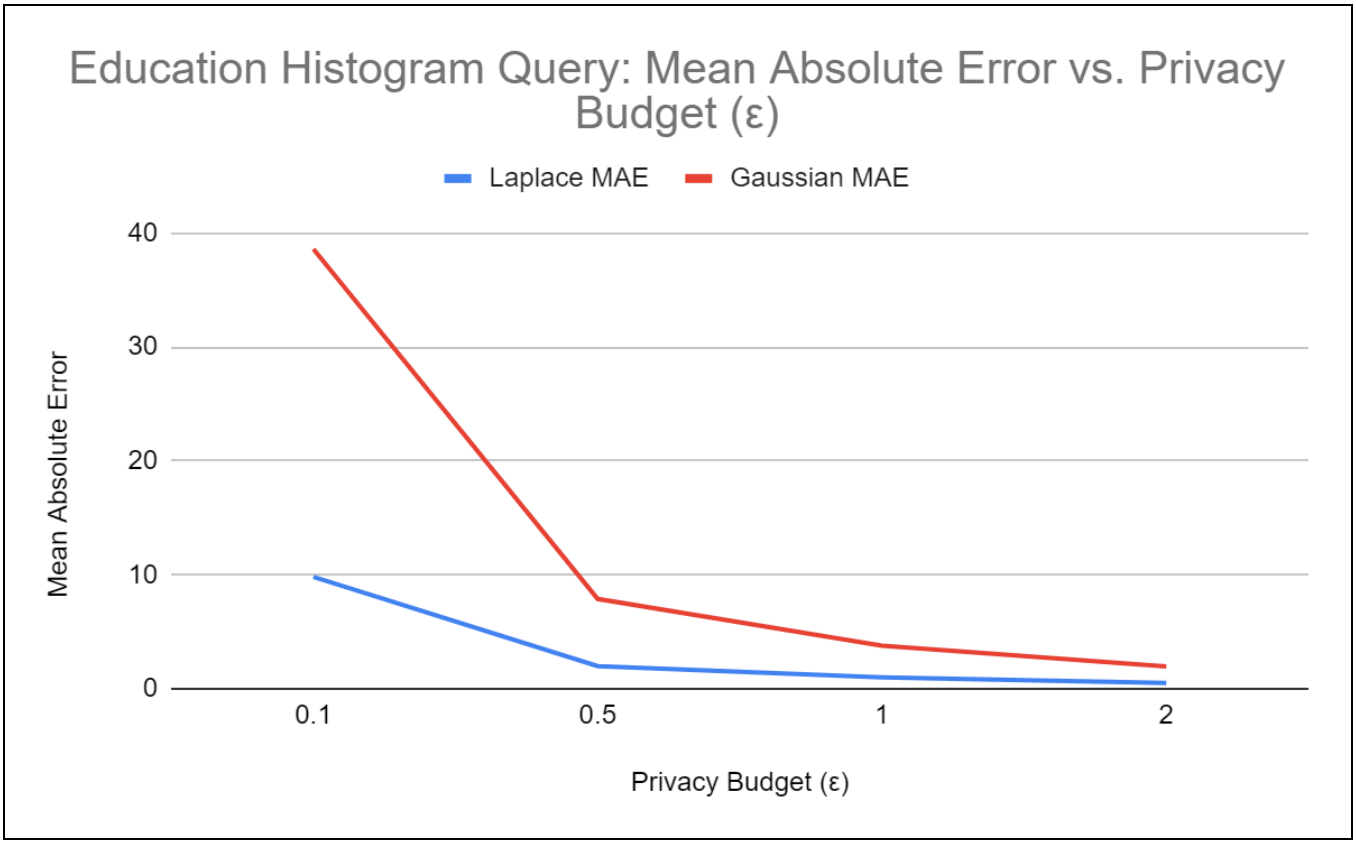

Figure 3. Mean absolute error for the education histogram query under the Laplace and Gaussian mechanisms across different privacy budgets. As ε increases, the error decreases for both mechanisms. The Laplace mechanism consistently produces lower error than the Gaussian mechanism, and the gap is larger than in the mean query because noise is added across multiple histogram bins.

3.7.4 Overall Comparison

For all three types of queries, the Laplace mechanism outperformed the Gaussian mechanism in this implementation for all values of ε. The reasons for this may lie in the relatively low sensitivity of the chosen queries and in the direct optimization of the Laplace mechanism for this sensitivity. The Gaussian mechanism followed a similar trend in its performance, improving in accuracy as ε increased, but added more noise overall in this optimization. Based on this experiment, it can be concluded that for simple aggregate data such as counts, bounded means, and histograms, the Laplace mechanism provided a better trade-off between utility and privacy in this project.

Query	Mechanism	ε = 0.1 MAE	ε = 0.5 MAE	ε = 1.0 MAE	ε = 2.0 MAE
Count > $50K	Laplace	9.1136	1.8797	1.0359	0.565
Count > $50K	Gaussian	37.0213	7.1031	4.1219	2.0497
Mean Age	Laplace	0.0235	0.005	0.0025	0.0013
Mean Age	Gaussian	0.0981	0.0187	0.0098	0.0045

Education Histogram	Laplace	9.8272	1.9925	0.9931	0.5037
Education Histogram	Gaussian	38.6114	7.8826	3.7743	1.9694

Table 1. MAE results for Laplace and Gaussian mechanisms across all queries and privacy budgets.

4. State-of-the-Art and Well-Known Techniques

Several methods and concepts in differential privacy are directly relevant to this project.

Laplace Mechanism: This is one of the basic methods in differential privacy and is usually the first mechanism to apply for numeric aggregate queries **[1]**.

Gaussian Mechanism: This is a common mechanism for providing differential privacy and is especially useful in modern privacy-preserving applications, especially in scenarios where there is a high probability of repeated use **[3], [4]**.

Smooth Sensitivity: Global sensitivity is sometimes too conservative, especially for certain aggregate queries such as median-like aggregate queries or for queries that are affected by outliers **[2]**.

Rényi Differential Privacy: RDP is a framework for privacy analysis that is especially useful in scenarios where the Gaussian mechanism is repeated **[4]**.

Practical DP: Robust practical implementations of differential privacy are facilitated by established frameworks, such as OpenDP and Google's differential privacy libraries **[6], [7]**. These resources offer a standardized suite of tools designed to streamline the integration of privacy-preserving mechanisms into data analytics workflows. By providing a consistent methodological approach, these libraries ensure that differential privacy is implemented both reliably and effectively across various applications **[5], [8]**.

In the context of this project, the Laplace and Gaussian mechanisms are the basic methods being directly considered, and smooth sensitivity and RDP, and DP tools, are the overall state of the art **[1]–[8]**.

5. Conclusion

This project evaluated the Laplace and Gaussian differential privacy mechanisms using count, mean, and histogram queries on the Adult Income dataset, varying the privacy parameter ε. Results confirmed the privacy-utility trade-off: increasing ε improved accuracy. Critically, the Laplace mechanism consistently outperformed the Gaussian mechanism across all queries and ε values, with the largest difference seen in the count and histogram queries. For low-sensitivity queries like those tested, Laplace is the better choice, although Gaussian may be necessary for approximate (ε, δ)-DP and composition. Ultimately, the performance of differential privacy depends crucially on the choice of mechanism and its parameters.

6. References

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